



Python for Fixed Income

COURSE 3 · MARKET DATA AND TIME SERIES

Market Calendars and Cleaning

WEEK 1 · DAY 3

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Which days did the market have?

The market's days are the dates the data has. No holiday list decides them.

missing_weekdays the weekdays a series has no row for

align series side by side, no days invented

stale_runs a value repeated day after day

A holiday is a gap, never a zero change. A missing day is never filled for a calculation.

Today's running data

Treasury par yields: 2,940 market days, 2015 to 2026, in percent. Mostly the 10 Yr.

course3_synthetic_spreads.csv: invented for this course, in bp. Not ICE, Moody's, or any index.

Calendar days, weekdays, market days

4,292

calendar days: `pd.date_range`

3,066

weekdays: `pd.bdate_range`

2,940

market days: the index

126

weekdays with no row

Gap to the row before	Rows
1 day	2,295
3 days (Friday to Monday)	518
2 days	31
4 days (a long weekend)	95

A change is from the previous row, whatever the gap. The Tuesday after a Monday holiday is a change from Friday.

Mondays have the fewest rows: 538.

The file as its own calendar

```
def missing_weekdays(index):  
    _check_index(index, "index")  
    # every Monday to Friday in the span  
    weekdays = pd.bdate_range(index[0], index[-1])  
    # the ones with no row  
    return weekdays.difference(index)
```

_check_index stops on

an index that is not dates

a date that appears twice

dates not in ascending order

126

weekdays with no row in the Treasury file

2026-09-07

the last one: a Monday

**No holiday calendar is used: the answer is every weekday
the data skipped, for whatever reason.**

A holiday calendar is not the market calendar

USFederalHolidayCalendar against the file	Count
Federal holidays in the span	121
Missing weekdays that are federal holidays	117
Missing weekdays that are Good Fridays	8
Missing and neither (2018-12-05)	1
Federal holidays with a row (Fridays)	4
Good Fridays with a row	4

Federal holidays with a row	
2017-11-10	Veterans Day
2021-06-18	Juneteenth
2021-12-31	New Year's Day
2023-11-10	Veterans Day

No reason is given for any of these days. A holiday calendar is a guess at the record; the file is the record.

FRED's dots and Treasury's absent rows

Treasury's file

No curve that day: no row at all.

```
2026-09-04, ... 2026-09-08, ...
```

FRED's API

No value that day: a row whose value is a dot.

```
{"date": "2026-09-07", "value": "."}
```

The FRED-format sample: 43 rows, one NaN, on 2026-09-07, a weekday the file has no row for.

A live cell, two ways

No key (the saved notebook):

Live cell skipped: no FRED key was found.

With your key:

DGS10 since 2015: every dot is one of the file's missing weekdays

Joining two series: align

how="inner" only the dates every series has

how="outer" every date any series has; NaN where a series has no row

Join	Rows	What it shows
File and FRED sample, outer, dots kept	43	a row on 2026-09-07, NaN in both
File and FRED sample, inner, dots dropped	42	one row per market day, equal
10 Yr and 4 Mo, inner	989	from 2022-10-19, when both exist
10 Yr and 4 Mo, outer	2,940	1,951 NaN in 4 Mo before it

An outer join puts NaN in the gaps, and can bring in a date the market never had. align never fills.

Forward fill: for display only

Date	10 Yr	Filled	Change bp	Filled bp
2026-09-04	4.78	4.78	1.0	1.0
2026-09-07	NaN	4.78	NaN	0.0
2026-09-08	4.80	4.80	2.0	2.0

Over the span	Market	Filled
Changes	41	42
Zero changes	3	4
Mean absolute, bp	3.976	3.881
Standard deviation, bp	4.909	4.854

ffill put Friday's 10 Yr on Labor Day: a change of 0.0 bp on a day with no market.

Never filled for a calculation. Forward fill is for display: on a copy, labelled, never before a change, a window, or a join.

Stale values: stale_runs

```
# a new run starts where the value changes
run_id = (series != series.shift()).cumsum()
for _, run in series.groupby(run_id): ...
```

One row per run of at least min_length equal values: start, end, length, value. Drop gaps first.

Runs	3 or more	5 or more	2 or more
ig_spread_bp (synthetic)	4	0	107
hy_spread_bp (synthetic)	0	0	29
Treasury 2 Yr	84		

A run is a reason to look

Real data repeats: the 2 Yr's longest run is 8 rows.
stale_runs says where; another source, or the person who sends the file, says whether.

Four new tests; 13 passed.

Excel side by side

Excel (tested)	Returns
=WEEKDAY(DATE(2026,9,4),2)	5 (Friday)
=WEEKDAY(DATE(2026,9,6))	1: Sunday is 1
=NETWORKDAYS(A2,A2941)-COUNT(A2:A2941)	126
=NETWORKDAYS(DATE(2026,1,1),DATE(2026,10,2))	197
=NETWORKDAYS(DATE(2026,1,1),DATE(2026,10,2),H2:H8)	190 = 2026 rows
=WORKDAY(DATE(2026,9,4),1)	2026-09-07
=WORKDAY(DATE(2026,9,4),1,H2:H8)	2026-09-08
=XLOOKUP(DATE(2026,9,7),A2:A2941,B2:B2941)	#N/A
=AND(C4=C3,C3=C2), filled; COUNTIF TRUE	4

Tested in Excel 16.0 build 20430, 2026-10-04. H2:H8 holds the missing weekdays of 2026.

Save the day: amend a message

```
git commit -q -m "Add missing_weekdays, align, and stale_run"  
git commit --amend -q -m "Add missing_weekdays, align, and stale_runs, with four tests"  
git log --oneline      # one commit, a new short hash
```

1. Spot it

stale_run: the function is stale_runs, and the tests are not named.

2. Amend

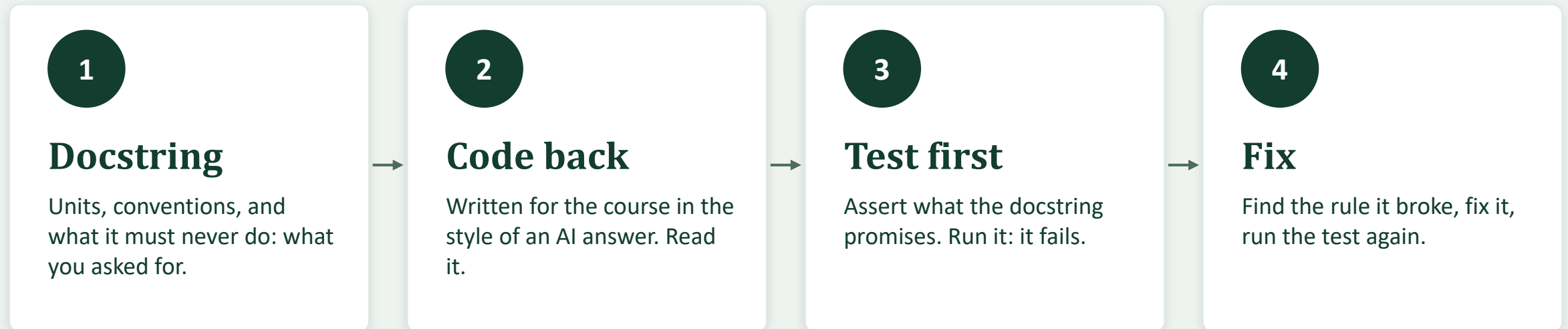
--amend replaces the last commit: same files, new message, new hash.

3. Only yours

Amend only what nobody else has. Pushed? Fix with a new commit.

The AI check

The docstring is the prompt. The code that comes back runs, and has one data bug.



Today: `closed_days` builds the closed days from the federal holiday calendar: 121 dates, not 126, and no Good Friday.

Code that runs is not the same as code that is right.

What you can do now

- Count calendar days, weekdays, and market days; find missing weekdays
- Say why a holiday calendar is not the market calendar
- Join series with align, inner or outer, and never fill a gap
- Keep forward fill for display; find stale runs, and look
- Count market days in Excel with a holiday list from the data

Week 1, Day 4 Resampling and the as-of value: which observation stands for a week or a month